Statistics

Chapter 9 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 13

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\mu = \text{Mean} = 8.6$$
$$\sigma = \text{Standard deviation} = 2.0$$
$$n = \text{Sample size} = 35$$
$$\overline{x} = 8.8$$

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{8.6 - 8.8}{2.0/\sqrt{35}} \approx -0.59$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(Z < 0.59)$ is given in the row starting with 0.5 and in the column starting with .09 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \geq 8.8) &= P(Z > -0.59) \\ &= P(Z < 0.59) \\ &= P(Z < 0) + P(0 < Z < 0.59) \\ &= 0.5000 + 0.2224 \\ &= 0.7224 \\ &= 72.24\% \end{aligned}$$

Result

$$0.7224 = 72.24\%$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\mu = \text{Mean} = 1210$$
$$\sigma = \text{Standard deviation} = 325$$
$$n = \text{Sample size} = 25$$
$$x = 1100$$

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{1100 - 1210}{325/\sqrt{25}} \approx -1.69$$

Determine the corresponding probability using the normal probability table in the appendix. $P(Z < 1.69)$ is given in the row starting with 1.6 and in the column starting with .09 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \leq 1100) &= P(Z < -1.69) \\ &= P(Z > 1.69) \\ &= 1 - P(Z < 1.69) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.69) \\ &= 1 - 0.5000 - 0.4545 \\ &= \end{aligned}$$

Result

$$0.0455 = 4.55\%$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 160\\ s &= \text{Sample standard deviation} = 20\\ n &= \text{Sample size} = 40\\ c &= \text{Confidence level} = 98\% = 0.98\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 40 - 1 = 39$ and in the column with $c = 98\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.426$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.426 \times \frac{20}{\sqrt{40}}\\ &\approx 7.6717\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 160 - 7.6717 = 152.3283$$

$$\overline{x} + E = 160 + 7.6717 = 167.6717$$

We are 98% confident that the true population mean number of customers per day is between 152.3283 and 167.6717.

Result

$$(152.3283, 167.6717)$$

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PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 2.65\\ s &= \text{Sample standard deviation} = 0.45\\ n &= \text{Sample size} = 20\\ c &= \text{Confidence level} = 90\% = 0.90\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 20 - 1 = 19$ and in the column with $c = 90\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.729$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 1.729 \times \frac{0.45}{\sqrt{20}}\\ &\approx 0.1740\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 2.65 - 0.1740 = 2.4760$$

$$\overline{x} + E = 2.65 + 0.1740 = 2.8240$$

We are 90% confident that the true population mean waiting time is between 2.4760 minutes and 2.8240 minutes.

Result

$$(2.4760, 2.8240)$$

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Solution A

Step 1

Given:

$$n = \text{Sample size} = 6$$
$$c = \text{Confidence level} = 95\% = 0.95$$

Confidence interval

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{826 + 931 + 1126 + 918 + 1011 + 1101}{6} = \frac{5913}{6} = 985.5$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(826 - 985.5)^2 + (931 - 985.5)^2 + (1126 - 985.5)^2}{(981 - 985.5)^2 + (1011 - 985.5)^2 + (1101 - 985.5)^2}{6 - 1}} \approx 115.4968$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 6 - 1 = 5$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.571$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 2.571 \times \frac{115.4968}{\sqrt{6}} \\ &\approx 121.2262 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 985.5 - 121.2262 = 864.2738$$

$$\overline{x} + E = 985.5 + 121.2262 = 1106.7262$$

We are 95% confident that the true population mean number of copies per machine is between 864.2738 thousand and 1106.7262 thousand.

🎯 Result

$$(864.2738, 1106.7262)$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$n = \text{Sample size} = 5$$
$$c = \text{Confidence level} = 95\% = 0.95 \quad \text{(Assumption)}$$

We note that the sample of the last 5 callers is not a random sample from the population and thus it is likely that the sample is not representative for the population. This then implies that the confidence interval won't be appropriate to calculate as it won't reflect the scope of the true population mean.

Next, we also note that the data values were rounded to the nearest half hour, as the parents didn't measure the number of hours of watching TV precisely. The confidence interval is then not appropriate to be calculated, as the data values won't accurately reflect the population.

Step 2

For those who need to calculate the confidence interval anyway or those who decide otherwise, you find the calculation of the confidence interval below:

Confidence interval

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{3.0 + 3.5 + 4.0 + 4.5 + 3.0}{5} = \frac{18.0}{5} = 3.6$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(3.0 - 3.6)^2 + (3.5 - 3.6)^2 + (4.0 - 3.6)^2 \\ (4.5 - 3.6)^2 + (3.0 - 3.6)^2}{5 - 1}} \approx 0.6519$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 5 - 1 = 4$ and in the column with $c = 95\%$ in the table of the Students T distribution:

$$t_{\alpha/2} = 2.776$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 2.776 \times \frac{0.6519}{\sqrt{5}} \\ &\approx 0.8093 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 3.6 - 0.8093 = 2.7907$$

$$\overline{x} + E = 3.6 + 0.8093 = 4.4093$$

We are 95% confident that the true population mean number of hours of TV watched last night is between 2.7907 hours and 4.4093 hours.

Result

Not appropriate

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 240\\ s &= \text{Sample standard deviation} = 35\\ n &= \text{Sample size} = 16\\ c &= \text{Confidence level} = 95\% = 0.95 \qquad \text{(Assumption)}\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 16 - 1 = 15$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.131$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.131 \times \frac{35}{\sqrt{16}}\\ &= 18.64625\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 240 - 18.64625 = 221.35375$$

$$\overline{x} + E = 240 + 18.64625 = 258.64625$$

We are 90% confident that the true population mean number of widgets produced per day is between 221.35375 and 258.64625.

Since 250 is between 221.35375 and 258.64625, it is likely that the population mean is 250 when the mean daily widget production did not increase and thus it is not reasonable to conclude that the mean daily widget production has increased.

Result

Not reasonable

Chapter 9 Exercise 8

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PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\sigma = \text{Population standard deviation} = 0.90$$
$$E = \text{Margin of error} = 0.10$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 0.90}{0.10}\right)^2 \approx 312$$

Thus we require a sample of size at least 312.

Result

$$n = 312$$

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PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\begin{aligned} \text{Range} &= 5.00 \text{ to } 155.00 \\ E &= \text{Margin of error} = 4.00 \\ c &= \text{Confidence coefficient} = 95\% = 0.95 \end{aligned}$$

We know that nearly all data values are within 2 standard deviations of the mean (when considering a normal distribution). Since the range contains all data values, the range is about 4 standard deviations and thus we estimate the standard deviation as about one fourth of the range.

$$\sigma \approx \frac{\text{Range}}{4} = \frac{155.00 - 5.00}{4} = \frac{150.00}{4} = 37.5$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 37.5}{4.00}\right)^2 \approx 338$$

Thus we require a sample of size at least 338.

Result

$$n = 338$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$x = \text{Number of successes} = 120$$
$$n = \text{Sample size} = 200$$
$$c = \text{Confidence level} = 95\% = 0.95$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{120}{200} = 0.6$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.6(1-0.6)}{200}} \\ &\approx 0.0679 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.6 - 0.0679 = 0.5321$$

$$\pi + E = 0.6 + 0.0679 = 0.6679$$

We are 95% confident that the true population proportion of residents who believe the tax rate is too high is between 0.5321 and 0.6679.

We note that 0.50 does not lie in the confidence interval, while the confidence interval contains only values above 0.50. This then implies that we expect more than 50% of the taxpayers to feel that the taxes are too high and thus it is reasonable to conclude that the majority of the taxpayers feel that the taxes are too high.

Result

Reasonable

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PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 8\% = 0.08$$
$$E = \text{Margin of error} = 2\% = 0.02$$
$$c = \text{Confidence coefficient} = 98\% = 0.98$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{2.33^2 \times 0.08(1-0.08)}{0.02^2} \approx 999$$

Thus we require a sample of size at least 999.

Result

$$n = 999$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\pi = \text{Sample proportion} = 25\% = 0.25$$
$$n = \text{Sample size} = 500$$
$$c = \text{Confidence level} = 98\% = 0.98$$

Confidence interval

For confidence level $1 - \alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 2.33 \times \sqrt{\frac{0.25(1-0.25)}{500}} \\ &\approx 0.0451 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.25 - 0.0451 = 0.2049$$

$$\pi + E = 0.25 + 0.0451 = 0.2951$$

We are 98% confident that the true population proportion of adults who currently smoke is between 0.2049 and 0.2951.

We note that the confidence interval does not contain 0.30 (30%), while the confidence interval contains only values below 0.30. This then implies that we would agree that the proportion is less than 30 percent.

🎯 Result

$$(0.2049, 0.2951)$$

PROBLEM STATEMENT

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Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 40\% = 0.40$$
$$E = \text{Margin of error} = 3\% = 0.03$$
$$c = \text{Confidence coefficient} = 98\% = 0.98$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{2.33^2 \times 0.40(1-0.40)}{0.03^2} \approx 1448$$

Thus we require a sample of size at least 1448.

Result

$$n = 1448$$